Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Deok-Ho Kim	Email	tipa@jhu.edu
Institute Name	Johns Hopkins University School of Medicine			
Address	Rm. 724, Ross Research Building 720 Rutland Avenue Baltimore, MD 21205			

2. Technology Overview

Title of Proposed Technology	Advanced Human Aged Engineered Muscle Tissue Platform for Preclinical Sarcopenia Drug Discovery			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio·Medical		Bio·Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Regenerative & Gene-derived Therapeutics	Medical Bio	Small-Molecule Pharmaceuticals
Concept of Proposed Technology	Sarcopenia is a progressive, age-associated degenerative disorder characterized by a decline in skeletal muscle mass, contractile strength, and metabolic capacity. Despite its growing clinical importance, preclinical drug discovery for sarcopenia remains fundamentally limited by the lack of human-relevant experimental platforms that faithfully recapitulate age-dependent muscle dysfunction. The research institution participating in this joint program possesses a unique and advanced capability to fabricate three-dimensional human aged engineered functional skeletal muscle tissues, which reproduce key hallmarks of sarcopenic pathology at functional, metabolic, and molecular levels. In addition, the platform is designed to flexibly integrate complementary preclinical systems and analytical technologies depending on the therapeutic modality of participating companies.			

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	In contrast, current preclinical pipelines predominantly rely on rodent models or simplified two-dimensional (2D) myotube cultures, which fail to capture critical features of human sarcopenia, including: ✓ Age-associated reduction in contractile force and fatigue resistance ✓ Impaired excitation–contraction coupling and calcium handling ✓ Mitochondrial dysfunction and metabolic inflexibility ✓ Dysregulated regenerative capacity and stress response pathways As a result, existing models exhibit limited translational relevance and low predictive power for clinical efficacy in human sarcopenia. The objective of this international joint **R&D project is to leverage the overseas institution's human aged engineered skeletal muscle tissue platform as a preclinical testbed for evaluating the efficacy and mechanisms of action of therapeutic candidates provided by Korean companies.** Through this collaboration, participating companies will be able to generate human-relevant functional, metabolic, and multi-omics datasets that support translational decision-making and de-risk subsequent clinical development.
Rationale for Joint R&D Project	Sarcopenia drug discovery requires experimental platforms that can accurately model age-dependent human skeletal muscle dysfunction and provide quantitative functional and mechanistic readouts. However, **human sarcopenia testing platforms are not currently available at the Korea, representing a critical bottleneck for domestic preclinical development.** While Korean SMEs and biotech companies possess diverse therapeutic candidates targeting muscle aging, mitochondrial dysfunction, and metabolic decline, they lack access to advanced human aged engineered skeletal muscle tissue systems capable of directly evaluating drug efficacy in physiologically relevant human contexts. In contrast, the partner research institution participating in this program has established a unique technological capability to generate three-dimensional engineered functional skeletal muscle tissues derived from aged human cells, along with integrated functional, metabolic, and multi-omics analysis pipelines. This platform enables high-content, quantitative assessment of human muscle contractility, bioenergetics, and molecular responses to therapeutic intervention. Therefore, this international joint R&D project is strategically designed to **bridge the gap between Korean therapeutic development capacity and overseas human sarcopenia modeling expertise.** By combining Korean companies' drug innovation pipelines with the overseas institution's world-class human muscle engineering platform, the project enables: ✓ Human-relevant preclinical efficacy evaluation ✓ Mechanism-informed translational biomarker discovery ✓ De-risking of clinical development for sarcopenia therapeutics ✓ Accelerated go/no-go decision-making for multiple industry participants

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	This collaboration represents a highly efficient and non-redundant international partnership, in which complementary strengths are integrated to establish a shared preclinical infrastructure for aging-related drug discovery that would not be achievable by either party independently.
Key Words	1. Sarcopenia 2. Human aged engineered muscle tissue 3. Preclinical drug discovery platform 4. Translational biomarker discovery 5. Regenerative medicine

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (6)
R&D Objectives	The overall objective of this international joint R&D project is to establish a human-relevant preclinical sarcopenia drug discovery platform by leveraging an overseas institution's advanced human aged engineered skeletal muscle tissue technology and integrating it with Korean companies' therapeutic development pipelines. 1. **Platform operationalization and validation** To operationalize and validate a human aged engineered skeletal muscle tissue platform for reproducible, high-content preclinical evaluation of sarcopenia therapeutics. 2. **Establishment of functional phenotyping assays** To implement quantitative assays for evaluating human muscle-specific functional outputs, including contractile force generation, fatigue resistance, and excitation–contraction coupling. 3. **Integration of metabolic and bioenergetic profiling** To establish standardized protocols for assessing mitochondrial function, ATP metabolism, and metabolic flexibility in human sarcopenic muscle tissues. 4. **Multi-omics data generation and biomarker discovery** To generate transcriptomic, proteomic, and metabolomic datasets that enable system-level characterization of therapeutic responses and identification of translational biomarkers. 5. **Preclinical efficacy evaluation of therapeutic candidates** To apply the platform to evaluate the efficacy and mechanisms of action of multiple sarcopenia drug candidates provided by participating Korean companies. 6. **Development of a shared translational dataset and decision framework** To create a unified data framework supporting cross-company comparison, go/no-go decision-making, and de-risking of subsequent IND-enabling and clinical development. **Technology Readiness Level (TRL)**			

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	● **Start of Project:** TRL 4 (Human engineered muscle tissue platform established and functionally validated in pilot studies) ● **End of Project:** TRL 6 (Human-relevant preclinical platform validated through multi-company drug testing and translational data generation)
R&D Target Specifications	**Platform Performance Targets** ● Reproducible fabrication of functional human aged muscle tissues with ≤20% inter-batch variability in baseline contractile force ● Stable maintenance of sarcopenic phenotypes for ≥28 days in culture ● Capability to process multiple therapeutic candidates (≥5 compounds) under standardized experimental conditions **Functional and Metabolic Readout Targets** ● Measurement of muscle contractile force with sensitivity sufficient to detect ≥15% functional improvement relative to untreated controls ● Quantification of mitochondrial respiration and ATP metabolism with consistent signal-to-noise ratios across experimental replicates ● Assessment of fatigue resistance and calcium dynamics with time-resolved functional resolution **Omics and Biomarker Output Targets** ● Generation of transcriptomic, proteomic, and/or metabolomic datasets for ≥3 therapeutic candidates ● Identification of drug-responsive molecular signatures associated with functional muscle recovery ● Establishment of cross-compound comparative biomarker panels supporting mechanism-based stratification **Translational and Industrial Impact Targets** ● Human-relevant preclinical datasets supporting go/no-go decision-making for participating companies ● Mechanism-informed efficacy profiles suitable for IND-enabling planning and partner engagement ● A reusable testing pipeline applicable to diverse therapeutic modalities (small molecules, biologics, nucleic acid-based therapies)
R&D Plan	**Year 1 (2026): Platform Setup and Baseline Characterization** ● Establishment of collaborative workflow with overseas institution and platform onboarding ● Implementation of human aged engineered skeletal muscle tissue fabrication and maturation protocols ● Baseline functional phenotyping including contractile force, fatigue resistance, and calcium dynamics ● Baseline metabolic profiling including mitochondrial respiration and ATP metabolism ● Establishment of preliminary quality attributes for tissue viability, functionality, and sarcopenic phenotypes **Year 2 (2027): Therapeutic Screening and Mechanistic Profiling** ● Selection and onboarding of therapeutic candidates from participating Korean companies ● Functional efficacy evaluation of drug candidates using contractility and fatigue assays ● Metabolic and bioenergetic assessment of therapeutic responses ● Multi-omics profiling (transcriptomics/proteomics/metabolomics) for mechanism analysis ● Refinement of quality attributes and performance benchmarks based on drug response

	data **Year 3 (2028): Therapeutic Screening and Mechanistic Profiling** ● Expansion of platform to multi drug testing ● Cross-compound comparative efficacy and mechanism analysis ● Translational biomarker discovery and validation ● Integration of functional, metabolic, and omics datasets into a shared analysis framework ● Finalization of operational protocols and generation of platform documentation

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	● Fabrication and maturation of human aged engineered skeletal muscle tissues ● Development and operation of functional phenotyping assays (contractility, fatigue, calcium dynamics) ● Metabolic and bioenergetic profiling (mitochondrial function, ATP metabolism) ● Multi-omics data generation and systems-level analysis ● Platform optimization and performance validation ● Technical guidance and knowledge transfer for platform operation ● Joint interpretation of experimental results and scientific publications
Expected R&R of Korean SME Partner	● Provision of therapeutic candidates for preclinical evaluation ● Design of treatment conditions and biological hypotheses ● Support of experimental execution and data interpretation ● Integration of platform-derived data into internal R&D pipelines ● Downstream translational planning (IND-enabling strategy, biomarker development) ● Intellectual property generation and development planning ● Participation in joint reporting and dissemination activities

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Technological Outcomes** ● Establishment of a human-relevant preclinical sarcopenia drug discovery platform based on aged engineered skeletal muscle tissues ● Generation of standardized functional, metabolic, and multi-omics datasets for human sarcopenic muscle phenotypes ● Validation of quantitative functional assays for evaluating therapeutic efficacy in human muscle tissues ● Development of a reusable experimental and data analysis framework applicable to diverse therapeutic modalities ● Creation of a sustained international collaboration model for human-based disease modeling **Expected Commercial Outcomes** ● Human-relevant preclinical efficacy datasets supporting go/no-go decision-making for Korean SMEs and

	biotech companies ● Reduction of early-stage clinical failure risk in sarcopenia drug development ● Acceleration of translational pipelines from preclinical testing to IND-enabling studies ● Strengthening of Korea's industrial competitiveness in aging-related and regenerative medicine markets

6. References

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